

Chapter 4

Reduced-Order Hydrodynamic Model and Trajectory Optimisation

The CFD studies in the CFD chapter resolve the flow around a single leg or the bare hull with high fidelity, but one unsteady case takes hours to days. Finding a good paddling trajectory needs the opposite trade-off: thousands of whole-robot evaluations, each of which only has to rank candidate gaits correctly. This chapter develops a reduced-order model for that purpose. A quasi-steady, per-link hydrodynamic force model is attached to a rigid-body simulation in MuJoCo [1], the joint trajectories are written as a low-dimensional parameter vector, and an evolution strategy (CMA-ES) searches that vector. The same pipeline is applied to two robots: an open toy quadruped with drag-type flippers, and the BODY2 robot with passively pitching lift-type flukes.

The chapter has four aims:

1. define the hydrodynamic and gait models precisely enough to reproduce them (Sections 4.2 and 4.3);
2. verify the implementation and the evaluation protocol, including three errors that were found and corrected during this work (Section 4.4);
3. show what the optimiser finds for a drag-type robot, and why its answer led to the lift-type fluke design (Section 4.5);
4. quantify the fluke design space of BODY2 with the uncalibrated model and state what must be calibrated before these numbers can be trusted (Sections 4.6 and 4.7).

All speeds in this chapter are model predictions with coefficients taken from the literature or from the hull CFD. None of them is a measured swimming speed.

4.1 Simulation platform and pipeline

Platform choice. The first implementation used Webots. BODY2’s legs are closed parallelogram and four-bar linkages. URDF cannot describe closed chains, so the loops had to be closed by hand after import. In Webots this produced two problems that could not be removed by parameter tuning. First, a closed four-bar has two assembly modes, and with all joints initialised at zero the solver placed different legs on different branches (“one leg always behaves differently”). Second, several robot instances in one world influenced each other’s assembly, so parallel comparison of gaits was unreliable. MuJoCo [1] was adopted instead. It simulates in generalised coordinates, closes loops with explicit equality constraints, and resets a state deterministically by overwriting `qpos/qvel`; identical parameters give bit-identical results (Section 4.4). MuJoCo has no free surface and no built-in fluid model suitable for partially immersed bodies. Both

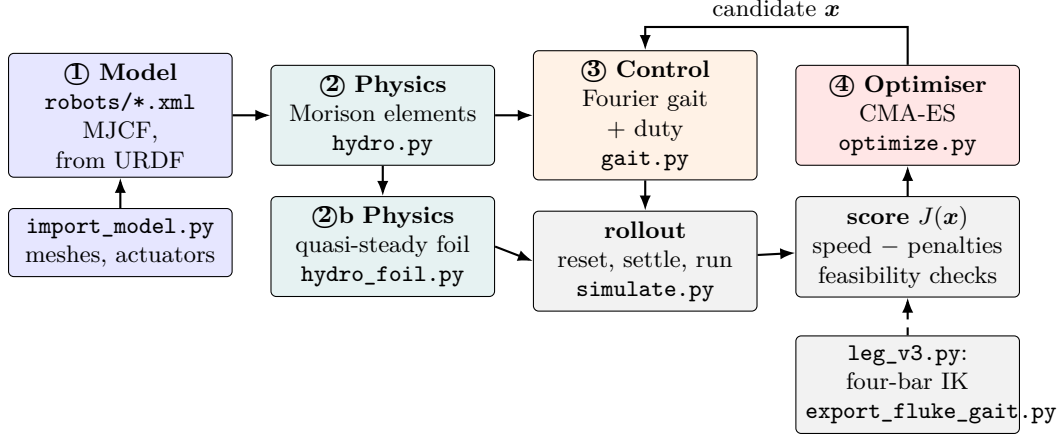


Figure 4.1: Structure of the `swimopt` pipeline. Only layer ① depends on the robot; hydrodynamic coefficients are assigned to links by name rules in a configuration file. The four-bar inverse kinematics of BODY2 enter only as feasibility checks and in the firmware export.

were therefore implemented as an external force module (Section 4.2), which also keeps every modelling assumption explicit.

Pipeline. The software (`swimopt`) is split into four layers so that a new robot changes only the first (Figure 4.1). The optimiser sees only a function $J(\mathbf{x})$ of a normalised parameter vector $\mathbf{x} \in [0, 1]^n$.

4.2 Hydrodynamic model

Two element types are used. Hull, links and drag-type paddles are *Morison elements*: forces depend on the local velocity of each collision geometry. The BODY2 flukes are *foil elements*: forces depend on angle of attack and follow a quasi-steady wing model. Both are applied as external wrenches (`xfrc_applied`) before every integration step.

4.2.1 Morison elements

For every immersed geometry i with volume V_i , orientation $\mathbf{R}_i$ and frontal areas $A_{i,k}$ normal to its local axes $k \in \{x, y, z\}$, the applied force is

$$\mathbf{F}_i = \underbrace{\rho g V_i f_i \mathbf{e}_z}_{\text{buoyancy}} - \underbrace{\frac{1}{2} \rho \mathbf{R}_i (\mathbf{C}_{d,i} \odot \mathbf{A}_i \odot \mathbf{v}_i^\ell \odot |\mathbf{v}_i^\ell|) f_i}_{\text{anisotropic quadratic drag}} - \underbrace{c_{v,i} \mathbf{v}_i f_i}_{\text{linear damping}} + \underbrace{m_{a,i} g \mathbf{e}_z}_{\text{added-mass compensation}}, \quad (4.1)$$

where $\mathbf{v}_i$ is the world velocity of the geometry centre, $\mathbf{v}_i^\ell = \mathbf{R}_i^\top \mathbf{v}_i$ its components in the body-fixed frame, $\odot$ the element-wise product, and $f_i \in [0, 1]$ the immersed fraction. The force acts at the geometry centre, so the wrench on the parent body also contains the moment $(\mathbf{p}_i - \mathbf{c}_b) \times \mathbf{F}_i$ about the body's centre of mass $\mathbf{c}_b$. This is the quasi-steady drag and inertia decomposition of Morison et al. [2], applied per link as is common for underwater vehicles [3] and used for paddling robots by Chen et al. [4, 5].

Anisotropy. Each element carries three drag coefficients. For a thin paddle the coefficient normal to its face is large and the two edge-wise coefficients are small (Table 4.1). This asymmetry is what allows a reciprocating paddle to produce net thrust at all; with an isotropic coefficient the force would always be exactly opposite to the paddle velocity.

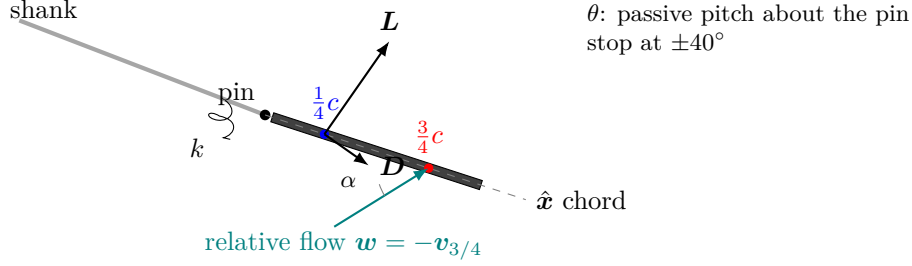


Figure 4.2: Foil element of the BODY2 fluke. The relative flow is taken at the three-quarter-chord point, the aerodynamic force acts at the quarter-chord point, and the pitching moment about the pin that drives the passive rotation θ follows from the lever arm between the two.

Partial immersion. The immersed fraction uses the true vertical half-extent h_i of the geometry at its current orientation, for example $h_i = \sum_k |R_{i,zk}| s_{i,k}$ for a box with half-sizes $s_{i,k}$ and $h_i = |R_{i,zz}| \ell_i + r_i$ for a capsule. With the water surface at z_w ,

$$f_i = \text{clip}\left(\frac{z_w - (p_{i,z} - h_i)}{2h_i}, 0, 1\right). \quad (4.2)$$

The linear profile is exact for the volume of an upright box and an approximation otherwise. Mesh geometries are replaced by their vertex bounding box scaled by a fill factor (default 0.55).

Added mass. An explicit term $-m_a \dot{\mathbf{v}}$ uses the acceleration of the current step and is numerically stiff for light links. Instead, the added mass $m_{a,i} = C_{a,i} \rho V_i$ is added to the inertial mass of the parent body once at initialisation, with the rotational inertia scaled by the same factor, and its weight is cancelled by the constant upward force in Equation (4.1). The body then has the correct weight and an increased inertia. This is unconditionally stable, but it makes the added mass isotropic and independent of immersion, whereas the physical added mass of a plate is large only normal to its face [6]. The consequence is discussed in Section 4.7.

4.2.2 Quasi-steady foil elements for the passive fluke

The BODY2 fluke is a thin plate on a pin at the end of the shank. It rotates freely about the pin against a leaf spring. Its thrust comes from the force normal to the relative flow, so a model based on angle of attack is needed. Each foil is a box with chord c along its local x axis, span b along y and normal $\hat{\mathbf{z}}$; $S = cb$ and $AR = b/c$ (Figure 4.2).

Following the quasi-steady reduction of Theodorsen's theory [7], the relative flow is evaluated at the three-quarter-chord point, where the translational and rotational velocity contributions combine: $\mathbf{w} = -\mathbf{v}_{3/4}$, with $w_x = \mathbf{w} \cdot \hat{\mathbf{x}}$ and $w_z = \mathbf{w} \cdot \hat{\mathbf{z}}$. The angle of attack and dynamic pressure are

$$\alpha = \text{atan2}(w_z, w_x), \quad q = \frac{1}{2} \rho (w_x^2 + w_z^2) S f. \quad (4.3)$$

Lift acts normal to the in-plane flow and drag along it, $\mathbf{F} = q(C_L \hat{\mathbf{l}} + C_D \hat{\mathbf{w}})$, applied at the quarter-chord point. The coefficients blend attached-flow thin-wing values with a post-stall flat-plate model:

$$C_L(\alpha) = [\sigma C_{L\alpha} + (1 - \sigma) C_N] \sin \alpha \cos \alpha, \quad (4.4)$$

$$C_D(\alpha) = \sigma \left(C_{D0} + \frac{(C_{L\alpha} \sin \alpha \cos \alpha)^2}{\pi e AR} \right) + (1 - \sigma) C_N \sin^2 \alpha, \quad (4.5)$$

$$\sigma(\alpha) = \text{clip}\left(\frac{1.6 \alpha_s - |\alpha|}{0.6 \alpha_s}, 0, 1\right), \quad C_{L\alpha} = \frac{2\pi AR}{AR + 2}. \quad (4.6)$$

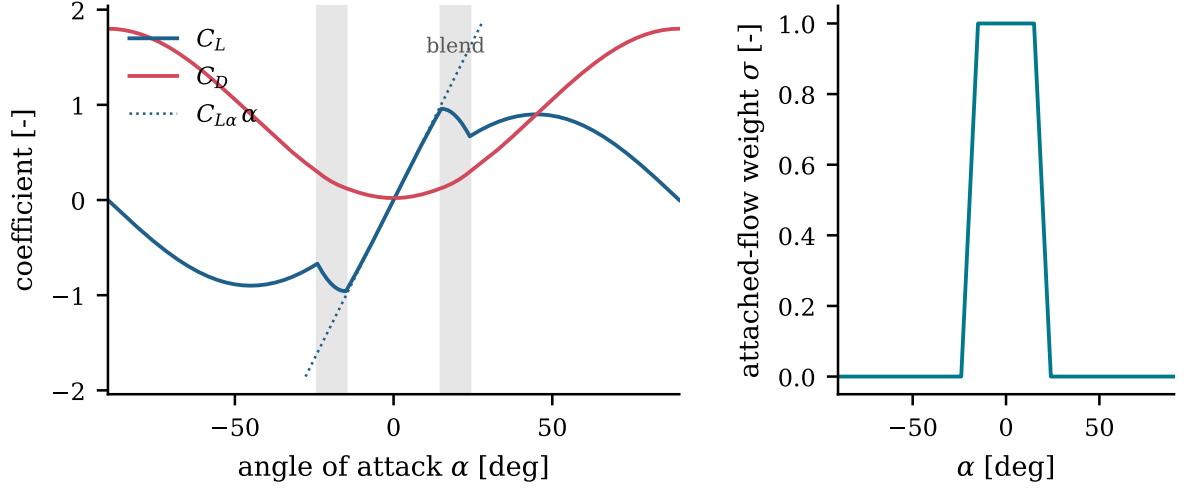


Figure 4.3: Foil coefficients for the BODY2 fluke ($AR = 3.13$, $C_{L\alpha} = 3.83/\text{rad}$, $\alpha_s = 15^\circ$, $C_N = 1.8$). Left: lift and drag coefficients; the grey bands are the blending ranges. Right: attached-flow weight σ .

Table 4.1: Hydrodynamic parameters used in this chapter. None has been calibrated against the physical robot yet, except the hull’s surge coefficients, which were fitted to the bare-hull CFD at 0.1 and 0.2 m/s (the CFD chapter).

robot	element (name rule)	$C_d(x, y, z)$	C_a	c_v [Ns/m]
toy	flipper (flip)	(2.2, 0.10, 0.10)	1.0	0.05
	leg link (link)	(0.5, 0.50, 0.25)	0.3	0.02
	trunk (trunk)	(0.9, 1.60, 1.80)	0.2	0.10
BODY2 fluke	thigh, shank (link)	(0.6, 0.6, 0.25)	0.3	0.02
	hull (trunk)	(0.093, 1.2, 1.5)	0.15	0.0154
	fluke foil (foil)	$c = 19.2 \text{ mm}$, $b = 60 \text{ mm}$, $\alpha_s = 15^\circ$, $C_{D0} = 0.02$, $e = 0.9$, $C_N = 1.8$, $C_a = 1.0$		
	passive hinge	$k = 2.5 \text{ mN.m/rad}$, range $\pm 40^\circ$		

The lift slope is Helmbold’s low-aspect-ratio correction [8]. The weight σ is one for $|\alpha| \leq \alpha_s$ and zero beyond $1.6\alpha_s$ (Figure 4.3). Buoyancy and added mass ($m_a = C_a \rho \pi (c/2)^2 b$) are handled as for Morison elements. The passive hinge is a MuJoCo joint with the leaf-spring stiffness k , a small damping, and a $\pm 40^\circ$ range limit.

4.3 Gait parameterisation and optimisation

4.3.1 Truncated Fourier series with duty-cycle warp

Every position actuator i follows

$$q_i(t) = o_i + r(t) \sum_{h=1}^H A_{i,h} \sin(h\vartheta(t) + \varphi_{i,h}), \quad (4.7)$$

with offset o_i , amplitudes $A_{i,h}$, phases $\varphi_{i,h}$ and a start-up ramp $r(t) = \min(1, t/1.5 \text{ s})$. The phase variable ϑ is warped so that a fraction d of the period (the power-stroke duty cycle) covers the first half-wave: with $\tau = (ft) \bmod 1$,

$$\vartheta = \begin{cases} \pi \tau / d, & \tau < d, \\ \pi + \pi(\tau - d)/(1 - d), & \tau \geq d. \end{cases} \quad (4.8)$$

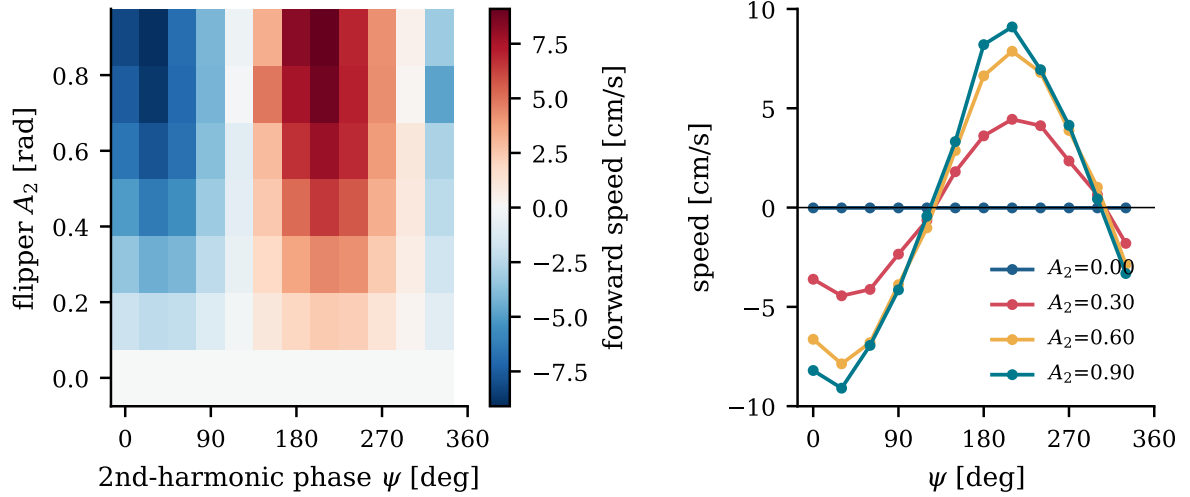


Figure 4.4: Toy robot, diagonal gait with all first harmonics fixed: forward speed as a function of the flipper’s second-harmonic amplitude A_2 and phase ψ . Left: map; right: sections. The row $A_2 = 0$ is zero everywhere, and $v(\psi + 180^\circ) = -v(\psi)$.

For $d = 0.5$ this is the ordinary sinusoid. Actuators can be grouped so that they share amplitude and offset (for example all hip joints), while phases stay individual. With n_u actuators, n_A amplitude groups and n_o offset groups, the parameter vector has

$$n = 2 + H(n_A + n_u) + n_o \quad (4.9)$$

entries (frequency, duty, amplitudes, phases, offsets), each mapped linearly from $[0, 1]$ to a physical range. For the toy robot ($n_u = 12$, three shared groups, $H = 2$) this gives $n = 35$; for BODY2 ($n_u = 4$, one group, $H = 1$, duty fixed) it gives $n = 8$, of which seven are optimised.

4.3.2 Why the flipper needs a second harmonic

For $H = 1$ every joint satisfies $q_i(t + T/2) - o_i = -(q_i(t) - o_i)$: the second half of each cycle is the first half reflected about the offset posture. For the toy robot, whose legs hang vertically and are fore-aft symmetric about their hips at zero offset, this reflects each leg’s stroke about the vertical, so quasi-steady forces from the two half-cycles cancel. A real paddle breaks the symmetry by feathering: it opens on the power stroke and closes on the recovery stroke, which is an effect at twice the stroke frequency. The term $A_{i,2} \sin(2\vartheta + \varphi_{i,2})$ is unchanged by the half-period shift and represents exactly this.

Figure 4.4 tests the argument on the toy robot with a fixed diagonal gait (hip, knee and flipper first harmonics fixed, zero offsets), sweeping only the flipper’s second-harmonic amplitude A_2 and phase ψ . Without the second harmonic the speed is below 0.2 mm/s (numerical noise of the settling transient). With it, the speed follows a sinusoid in ψ whose amplitude grows with A_2 up to 9.1 cm/s. Shifting ψ by 180° reverses the speed exactly (to 10^{-13} mm/s): the phase of the feathering relative to the stroke decides the swimming direction. A first-harmonic-only parameterisation would therefore search a space that contains no thrust at all for this robot.

Structured reduction. In the 35-dimensional toy space most phase combinations are not recognisable gaits. A reduced space fixes the inter-leg first-harmonic phases to a named pattern (diagonal, front-back, left-right, wave, in-phase) and couples the second-harmonic phases by $\varphi_{i,2} = \psi + 2\varphi_{i,1}$. The coupling gives every leg the same feathering timing relative to its own stroke. The remaining search variables are frequency, duty, three first- and three second-harmonic group amplitudes, ψ and three offsets: 12 dimensions.

4.3.3 Objective and feasibility

One evaluation (“rollout”) resets the robot, lets it settle in its initial posture for t_s (1.0–1.5 s), and then runs the gait for $T = 8$ s with $\Delta t = 2$ ms. The score is

$$J(\mathbf{x}) = \bar{u} - w_\psi \frac{|\Delta\psi|}{T} - w_E \bar{P} - w_\theta \text{std}(\theta_{\text{pitch}}) - w_z \text{std}(z) - w_{\dot{q}} \left(\frac{\max |\dot{q}_{\text{cmd}}|}{\dot{q}_{\text{max}}} - 1 \right)^+, \quad (4.10)$$

where $\bar{u}$ is the displacement along the initial heading divided by T , $\Delta\psi$ the net yaw change, $\bar{P}$ the mean absolute actuator power, and the standard deviations of trunk pitch and heave are taken over the second half of the run. The fitness is a ruler, not a result: its value depends on the weights and is not compared across settings. Only physical quantities are reported.

For BODY2 the objective includes $w_\theta = 0.5$, $w_z = 2.0$, $w_\psi = 0.3$ and a servo speed limit $\dot{q}_{\text{max}} = 14$ rad/s. Before a rollout, both extreme shank angles $q_r = o \pm A$ are checked against the four-bar inverse kinematics of the real leg (transmission angle $\geq 10^\circ$, knee angle $\geq 45^\circ$) and against the fluke–hull clearance limit $q_r \geq 15^\circ$. Infeasible candidates score -1 without being simulated. Rollouts whose trunk speed exceeds 5 m/s or whose joint speeds exceed 100 rad/s are aborted and score -1000 .

4.3.4 CMA-ES

The covariance matrix adaptation evolution strategy [9, 10] samples λ candidates $\mathbf{x}_k \sim \mathbf{m} + \sigma \mathcal{N}(\mathbf{0}, \mathbf{C})$, ranks them by J , moves the mean $\mathbf{m}$ to a weighted average of the best μ , and adapts the step size σ (cumulative step-size adaptation) and the covariance $\mathbf{C}$ (rank-one and rank- μ updates). Only ranks are used, so the scale of J does not matter, and no gradient is needed. The latter is essential here because contact with joint limits, feasibility checks and the divergence guard make J discontinuous. Box bounds $[0, 1]^n$ are handled by the `cma` package. Settings: $\lambda = 12$ for the full toy space and BODY2, $\lambda = 10$ for the 12-D reduced space, $\sigma_0 = 0.25$ (toy) and 0.3 (BODY2). A comparable CMA-ES workflow for gait selection of a fish-like robot is reported in [11]. The overall workflow (parameterise, simulate, optimise) follows Lee et al. [12]. They used a differentiable simulator and a gradient method; MuJoCo with the forces above is not differentiable, so a derivative-free optimiser is used.

4.4 Implementation and verification

Numerical checks. Table 4.2 summarises the checks. The vectorised force routine agrees with a per-geometry reference loop to $1.8e - 15$ N over 1500 steps of a moving gait and is 4.8 times faster (0.15 ms vs. 0.73 ms per step for 13 elements). This comparison exposed an error in the reference loop itself: a NumPy integer compared with a MuJoCo enum inside a Python `in` test evaluates false, so capsules were treated as spheres. The production path was not affected; the reference was corrected before the comparison. A second vectorised implementation of the foil routine, used for the parameter sweeps, agrees with the original to 10^{-14} in all reported metrics.

Flotation. With the legs held still, the time-averaged buoyancy after settling equals the weight to within 0.06 % (toy) and 0.03 % (BODY2) (Figure 4.5, left). The heave oscillation is lightly damped and still has a peak-to-peak amplitude of about 3 mm after 8 s. BODY2 settles with its reference point O2 at -6.4 mm and a bow-up trim of 1.3° . The target waterline from the hull design is -11 mm, so the provisional ballast (0.85 kg) is too light. This is one of the calibration items in Section 4.7.

Table 4.2: Verification checks of the implementation.

check	criterion	result
vectorised vs. reference forces	max. difference over 1500 steps	$1.8e - 15$ N
determinism	two rollouts, same $\mathbf{x}$	bit-identical
buoyancy balance	mean buoyancy / weight, legs still	1.0006 (toy), 1.0003 (BODY2)
null test	speed at zero stroke amplitude	-3.5 mm/s (toy), -0.12 mm/s (BODY2)
second-harmonic symmetry	$v(\psi) + v(\psi + 180^\circ)$	$< 10^{-13}$ mm/s
cost per rollout	9 s simulated, one CPU core	≈ 1.2 s (toy), ≈ 2.2 s (BODY2)

Virtual coast-down. The robot is launched at 0.25 m/s with the legs held at 40° , and $m_{\text{eff}}\dot{u} = -au^2 - bu$ is fitted to the simulated deceleration (Figure 4.5, right). The fit recovers $a = 1.93 \text{ N}\cdot\text{s}^2/\text{m}^2$ and $b = 0.144 \text{ N}\cdot\text{s}/\text{m}$ with $m_{\text{eff}} = 1.65$ kg (including added mass). The hull’s own coefficients, weighted by its immersed fraction (0.91), are $a = 0.24$ and $b = 0.014$. In the model the appendages, above all the flukes held at a large angle, therefore account for about 87 % of the quadratic and 90 % of the linear surge resistance at rest posture. Hull CFD alone would underestimate the robot’s drag, and the physical coast-down test should be done with the legs attached.

Evaluation protocol. Three errors in the evaluation protocol were found during this work. All results in this chapter were computed after correcting them.

1. *Start-up step.* The settling phase commanded zero joint angles, and the gait started from its offset angles. The stiff toy servos then moved the light joints at more than 100 rad/s in the first step, which triggered the divergence guard. Of 24 random candidates, 16 were rejected for this reason. After commanding the offset posture during settling, none were.
2. *Settling stroke.* Even with a slow servo, moving the legs from zero to the offset posture during settling is itself a stroke. On BODY2 the resulting glide persisted into the scored interval. With zero stroke amplitude the robot still covered 1.6 cm/s . The joints are now placed directly in the start posture at reset. The same zero-amplitude *null test* then gives -0.12 mm/s (BODY2) and -3.5 mm/s (toy).
3. *Pitch metric.* The pitch penalty originally used the RMS of the pitch angle, which is dominated by the static trim (1.3°) rather than the nodding it was meant to measure ($0.1\text{--}0.4^\circ$). It now uses the standard deviation, like the heave term.

Errors 1 and 2 had affected all earlier results. The earlier BODY2 optimum of 3.2 cm/s contained about 1.6 cm/s of settling glide. The earlier toy result had in addition been obtained with joint ranges interpreted in degrees (MJCF default) instead of radians.

4.5 Case A: toy quadruped with drag-type flippers

The toy robot (Figure 4.6, left) has a $200 \times 100 \times 50$ mm trunk and four legs with hip, knee and wrist joints driven by position servos ($k_p = 60 \text{ N}\cdot\text{m}/\text{rad}$, $|\tau| \leq 12 \text{ N}\cdot\text{m}$), and a 52×36 mm flipper on each wrist. These actuators are far stronger than any servo of a robot this size. The toy is therefore used for methodological questions only: which parameterisation, which search space, and which propulsion mode the optimiser chooses. Its speeds are not a prediction for BODY2. All toy runs use $w_\psi = 1.0$ and no energy, pitch or heave penalty.

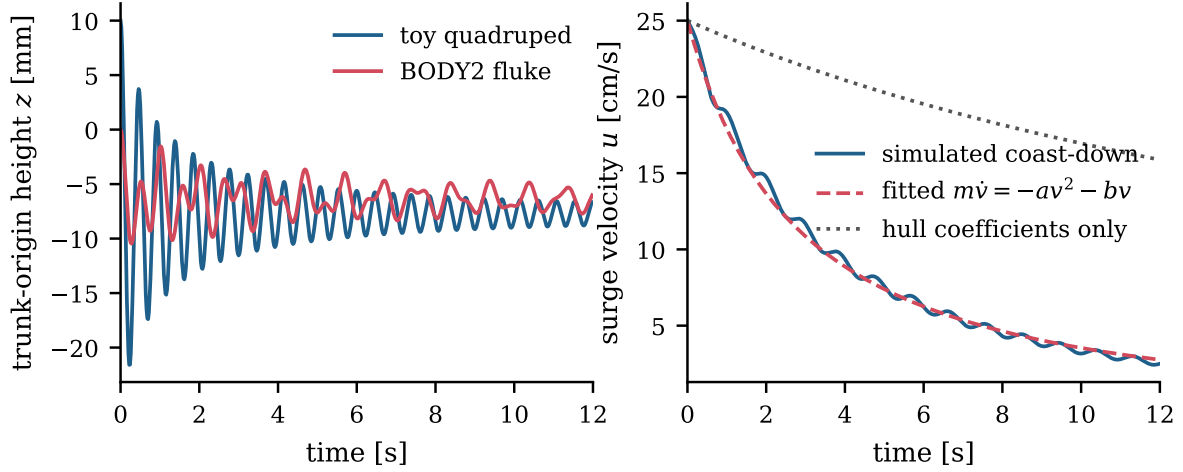


Figure 4.5: Left: settling from the design height with the legs held still. Right: simulated coast-down of BODY2 from 0.25 m/s, the fitted quadratic-plus-linear resistance, and the deceleration that the hull coefficients alone would give.

MuJoCo screenshot <code>shot_toy_quad.png</code> <code>python view.py config.json -demo</code> with "model": "robots/toy_quad.xml"; oblique side view, water plane visible	MuJoCo screenshot <code>shot_body2_fluke.png</code> <code>python view.py config_fluke.json</code> close-up of one leg: fixed thigh, shank, pin and fluke
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Figure 4.6: The two models. Left: toy quadruped, three pitch joints per leg (hip, knee, flipper wrist), 0.78 kg. Right: BODY2 with passive-pitch flukes, one actuated shank per leg, 1.35 kg including ballast.

4.5.1 Search space: full versus structured

Figure 4.7 compares CMA-ES on the full 35-dimensional space (two seeds, 480 rollouts each) with the structured 12-dimensional space (two seeds per pattern, 150 rollouts each). In the full space the best speeds were 12.5 cm/s and 4.9 cm/s for the two seeds. Both incumbents were found within 72 rollouts and never improved afterwards, and the step size did not contract ($\sigma \approx 0.18$ – 0.25 at the end). The search had not converged, and the two seeds disagree by a factor of 2.5. The structured space reached 20.2–22.2 cm/s with the diagonal pattern and 14.2–26.8 cm/s with the front-back pattern within 150 rollouts. Fixing the inter-leg phases and tying the feathering phase to the stroke removes directions that only rotate the robot or cancel thrust between legs, and this helps the optimiser more than the extra freedom of the full space.

4.5.2 Gait patterns

Figure 4.8 lists the best result per pattern. Diagonal (20.2–22.2 cm/s) and front-back (14.2–26.8 cm/s) are the fastest, followed by in-phase (15.0–16.4 cm/s), wave (5.8–9.8 cm/s) and left-right (2.0–8.6 cm/s). The seed-to-seed spread is as large as some differences between patterns, so with two seeds only the grouping “diagonal and front-back fast, wave and left-right slow” is supported. All of these gaits are violent: the trunk pitch standard deviation is 16–57°, and the servos deliver up to 58 W. Patterns that are symmetric about the sagittal plane (front-back, in-phase) have zero yaw by construction; the best diagonal gaits drifted by 2–13° in 8 s.

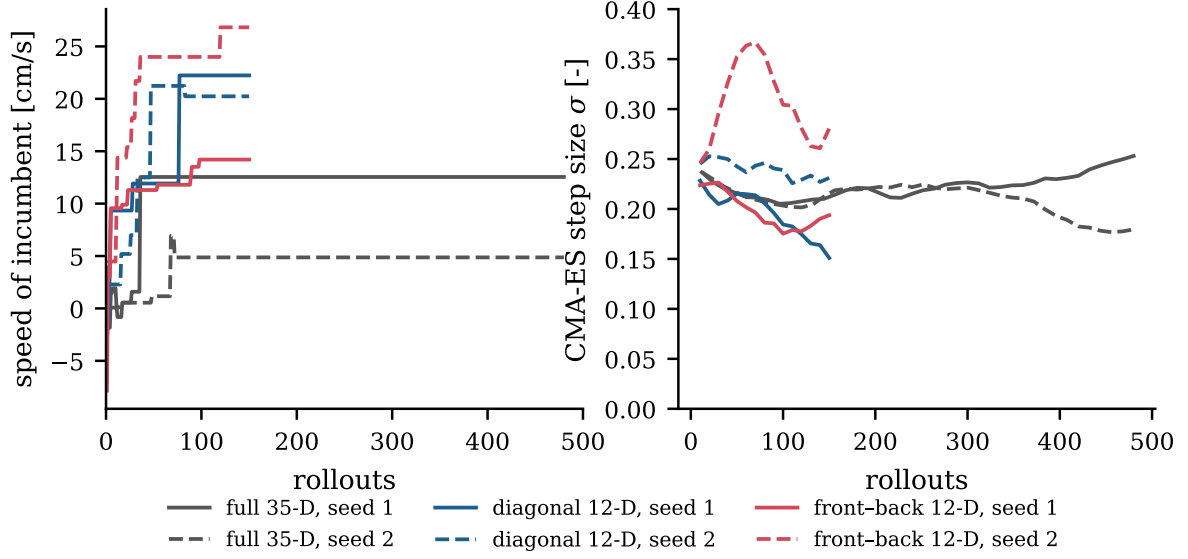


Figure 4.7: Toy robot. Left: speed of the best candidate found so far. Right: CMA-ES step size. Grey: full 35-D space; blue, red: structured 12-D space with diagonal and front-back phase patterns. Solid and dashed lines are two random seeds.

Duty cycle. Varying only the duty cycle of the best diagonal gait (optimum $d = 0.54$) leaves the speed between 21.6 cm/s and 22.4 cm/s over $d \in [0.35, 0.65]$. It halves at the ends of the range (Figure 4.9). The best front-back gait ($d = 0.56$) is fragile: five of the eleven duty values of the sweep lead to diverged rollouts. A robust optimum is preferable to a slightly faster one on the edge of instability, and the diagonal pattern is the better candidate.

4.5.3 The optimiser chooses lift-based thrust

The flipper forces of the optimised gaits were decomposed along the relative flow. With $\hat{\mathbf{w}} = -\mathbf{v}/|\mathbf{v}|$, the component $(\mathbf{F} \cdot \hat{\mathbf{w}})\hat{\mathbf{w}}$ is the drag part and the remainder the lift part. Their cycle-averaged forward components are shown in Figure 4.10. In all six structured optima the forward thrust comes entirely from the lift part (0.47–1.77 N for four flippers), while the drag part points backwards (−0.16– −1.07 N). This is the signature of lift-based propulsion in the sense of Fish [13]. It is not how a drag-based paddler such as the muskrat works, whose power stroke pushes water backwards [14]. The flippers sweep sideways and vertically as much as backwards (transverse path share 0.38–0.63).

Two points qualify the result. First, the thrust-weighted incidence is $58\text{--}71^\circ$, far above stall. The “lift” here is the flow-normal component of a flat-plate normal force, not attached-flow lift, which a Morison element cannot represent. Second, the two unconverged full-space results do not show the pattern (one is drag-dominated, the other produces almost no thrust). The result holds for the good gaits, not for every gait.

The same behaviour had been observed when running the toy model: all four legs swam in a lift-type manner. This was the reason for building BODY2’s lift-type fluke leg, and the reason for adding a foil element that represents attached-flow lift and stall explicitly (Section 4.2.2).

4.6 Case B: BODY2 with passive-pitch flukes

Model. The BODY2 model (Figure 4.6, right) consists of a $262 \times 116 \times 50$ mm hull (0.45 kg) with a low ballast block (0.85 kg) that places the centre of mass below the centre of buoyancy. Each leg has a fixed thigh at $\gamma = -85^\circ$, an actuated shank of 47 mm whose angle q_r is the

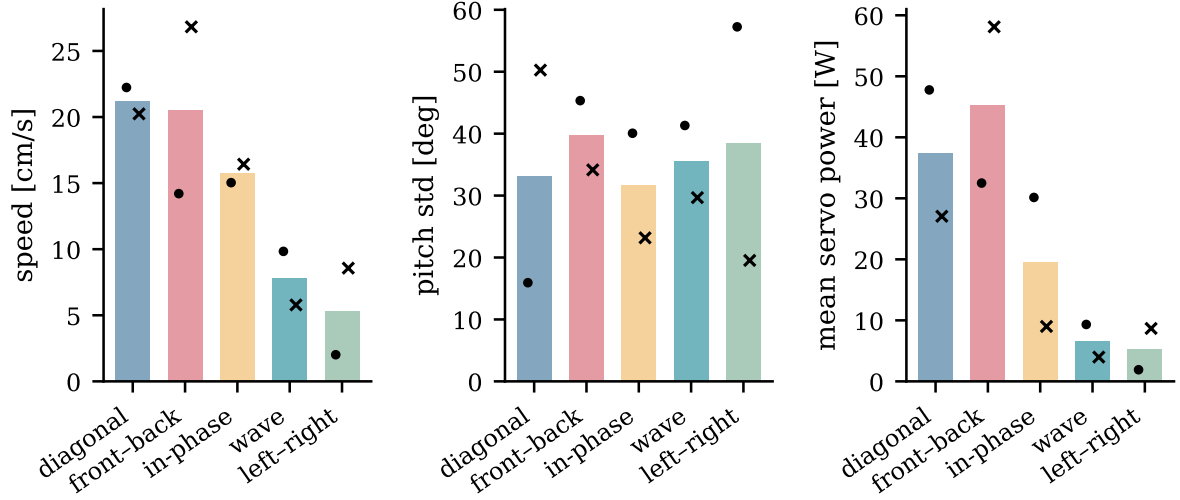


Figure 4.8: Toy robot, best gait per phase pattern after 150 rollouts in the 12-D space. Bars: mean of two seeds; dot and cross: seeds 1 and 2.

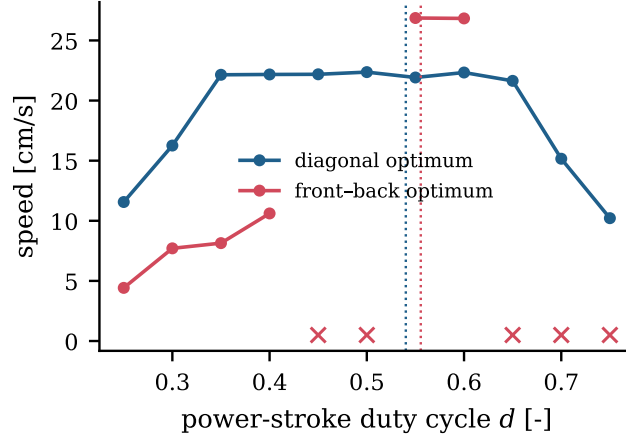


Figure 4.9: Speed versus power-stroke duty cycle for the best diagonal and front-back gaits, all other parameters fixed. Crosses: diverged rollouts. Dotted lines: optimised duty.

firmware’s rocker angle, and a fluke on a pin with a passive pitch joint (Table 4.1). The four-bar that drives the shank is not simulated; its inverse kinematics are used only for feasibility and for converting the result to servo angles. The shank servos are modelled as position actuators with $k_p = 3 \text{ N.m/rad}$ and $|\tau| \leq 0.8 \text{ N.m}$. The gait uses one harmonic per shank: the passive pitch joint supplies the stroke asymmetry that the second harmonic had to supply on the toy robot. As reference, the gait exported to the firmware from an earlier run is used: $f = 1.4 \text{ Hz}$, $q_r = 38.2^\circ \pm 23.1^\circ$, diagonal.

4.6.1 Phase pattern

At the reference frequency and amplitude, the diagonal pattern is the fastest (1.59 cm/s) and also by far the calmest: pitch standard deviation 0.12° and heave 0.3 mm (Figure 4.11). Front-back and in-phase strokes make the hull nod ($3.1\text{--}3.3^\circ$, 25 times more) and heave ($4.4\text{--}5.4 \text{ mm}$). This is consistent with the nodding reported from videos of the physical robot, whose hind legs currently stroke in phase. In the diagonal pattern the vertical fluke forces of the two diagonals cancel in pitch. The wave pattern turns the robot (47° in 8 s), and left-right swims backwards

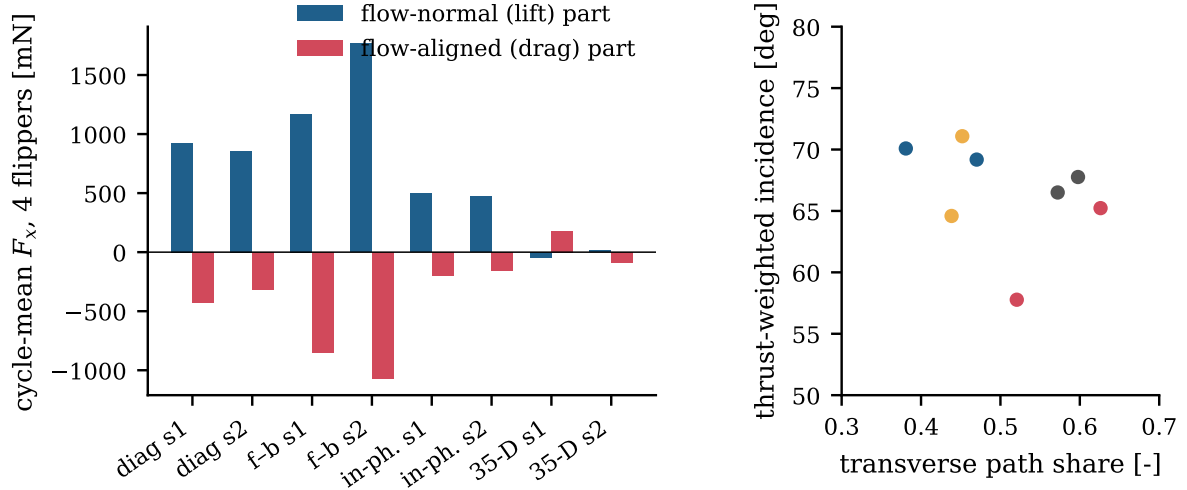


Figure 4.10: Left: cycle-mean forward force of the four flippers, split into the part normal to the relative flow (lift) and the part along it (drag), for the structured optima and the two full-space runs. Right: transverse share of the flipper path and thrust-weighted incidence (colours as in Figure 4.8; grey: full space).

with 17° roll. All further BODY2 results use the diagonal pattern.

4.6.2 Hinge stiffness

Figure 4.12 varies the leaf-spring stiffness. With a rigid fluke the robot does not move ($|\bar{u}| < 0.05$ cm/s). The passive pitch is what produces thrust. A very soft hinge lets the fluke swing onto its 40° stop, and thrust falls again. The optimum lies at $k \approx 2.5$ mN.m/rad for 1.4 Hz (the current spring) and at $k \approx 1$ mN.m/rad for 1.0 Hz. The optimal stiffness rises with frequency. This is consistent with the hydrodynamic moment on the fluke growing with the square of the stroke speed, so that a stiffer spring is needed for the same deflection. If the stroke frequency is changed, the spring should be changed with it.

4.6.3 Frequency and amplitude

Figure 4.13 maps speed over stroke frequency and shank amplitude. For amplitudes above 22.6° the mean shank angle is raised so that the lower extreme stays at the 15° clearance limit. Speed rises monotonically with amplitude, so the best stroke lies on the clearance constraint. Over frequency it peaks near 1.8 Hz. The best grid point, $f = 1.8$ Hz and $q_r = 43.5^\circ \pm 28^\circ$, scores 2.85 cm/s, against 1.59 cm/s for the reference stroke. Power grows faster than speed: going from the reference stroke to the best grid point raises the speed by a factor of 1.8 and the servo power by a factor of 2.3 (right panel).

Transient bias of the score. The score is the mean speed over the first 8 s of stroking and includes the acceleration from rest. The surge time constant of BODY2 is several seconds, so this underestimates the cruising speed. Rerunning both strokes for 30 s and fitting the last 10 s gives steady speeds of 3.28 cm/s (reference, 67 mW) and 4.99 cm/s (best grid point, 168 mW, 0.17 BL/s), i.e. 2.1 and 1.75 times the scored values. The ranking is unchanged. Absolute speeds should be quoted from long runs, and the scored interval should be lengthened or restricted to the last cycles in future searches.

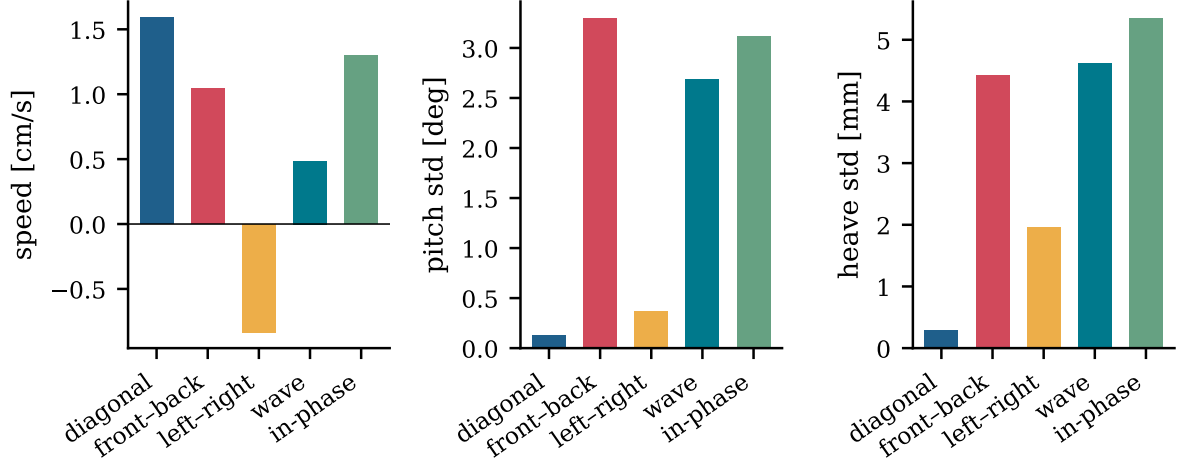


Figure 4.11: BODY2 at the reference stroke ($f = 1.4 \text{ Hz}$, $q_r = 38.2^\circ \pm 23.1^\circ$): speed, pitch standard deviation and heave standard deviation for five phase patterns.

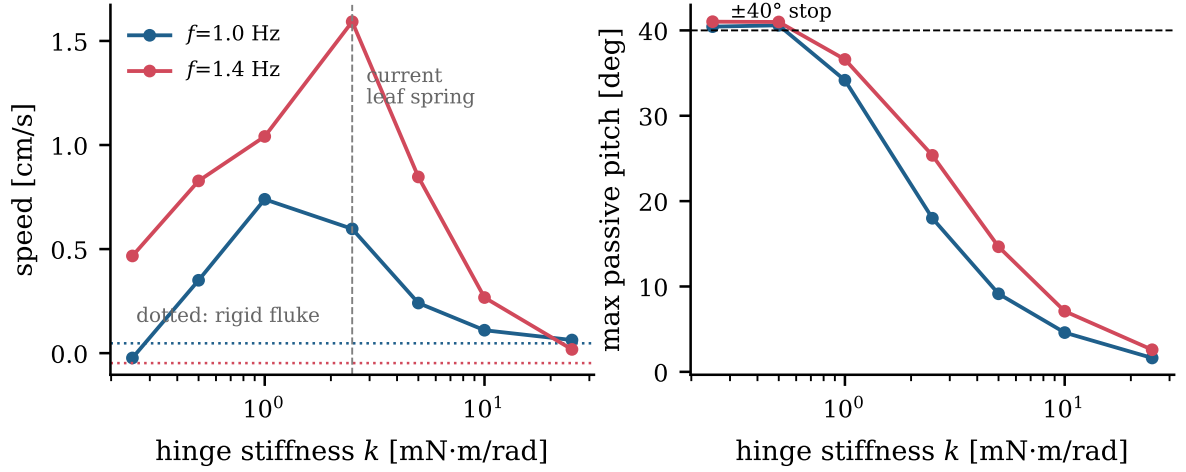


Figure 4.12: BODY2, diagonal reference stroke: speed (left) and maximum passive pitch (right) versus hinge stiffness at two stroke frequencies. Dotted lines: rigid fluke.

4.6.4 What the fluke actually does

Figure 4.14 shows two cycles of the best grid stroke after 6 s. The fluke pitches passively by $\pm 26^\circ$, lagging the shank. Its angle of attack still stays near $\pm 60\text{--}80^\circ$ for most of the stroke. The attached-flow branch of the foil model ($\sigma = 1$) is active in only 1.1% of the time, and the model is fully stalled in 98.2%. The reason is the low advance ratio: the three-quarter-chord point moves at up to 0.36 m/s while the robot cruises at 0.05 m/s, so the inflow arrives at about 82° to the stroke path. A passive pitch of more than 60° would be needed to bring α into the attached range, beyond the 40° stop. Each fluke produces a mean thrust of 3.5 mN but a mean upward force of 11 mN, three times larger. Flapping foils operate efficiently at moderate angles of attack and higher advance ratios [15]; at BODY2's speeds the fluke works as a feathering flat plate, and its thrust depends on the post-stall normal-force coefficient C_N , the least certain parameter of the model.

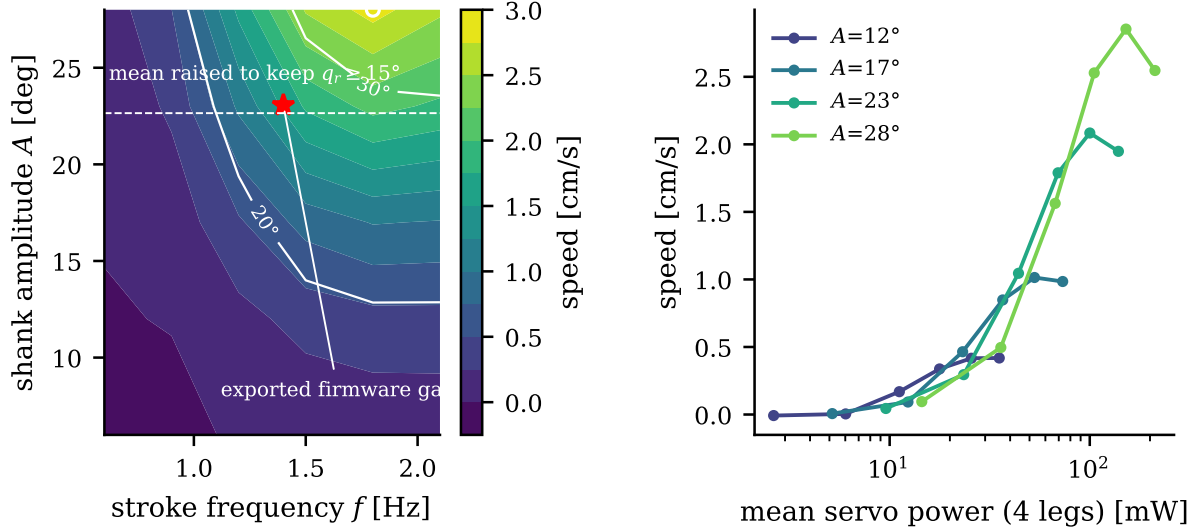


Figure 4.13: BODY2, diagonal pattern. Left: scored speed over frequency and shank amplitude; white contours: maximum passive pitch; star: reference firmware stroke; circle: best grid point. Above the dashed line the mean angle is raised to respect the clearance limit. Right: speed versus mean servo power for each amplitude.

Table 4.3: CMA-ES on BODY2, 144 rollouts per run. “Best” is the incumbent under Equation (4.10); “max.” is the fastest feasible candidate sampled.

constraint handling	seed	feasible	best: speed [cm/s]	max. speed [cm/s]
penalty (−1)	1	63 %	2.11	2.14
	2	73 %	0.02	1.27
	3	75 %	0.17	1.33
repair (shift mean)	1	95 %	1.30	2.28
	2	93 %	−0.01	3.03
grid, diagonal (Figure 4.13)	—	—	2.85	—

4.6.5 CMA-ES on BODY2

CMA-ES was run on the seven-dimensional BODY2 space (frequency, amplitude, offset, four phases) with the objective of Equation (4.10), three seeds and 144 rollouts each (Table 4.3). With infeasible candidates scored -1 , 25–37 % of all candidates were infeasible, and the incumbents reached at most 2.1 cm/s, below the best grid point. A repair step that shifts the mean shank angle into the feasible interval raised feasibility to 93–95 % and produced candidates up to 3.0 cm/s. The incumbents were still slow gaits, because at speeds of 1–3 cm/s the yaw and heave penalties are as large as the speed term: a gait with free phases that swims faster but turns by a few degrees scores worse than one that barely moves. The step size grew instead of shrinking in the repaired runs.

For BODY2 the useful sequence is therefore: fix the phases to the diagonal pattern, which is justified independently by Figure 4.11; handle the clearance constraint by repair; and rescale the penalties to the speed range of the robot. The remaining space (frequency, amplitude, offset, and optionally hinge stiffness) is small enough for the sweeps above to cover it directly.

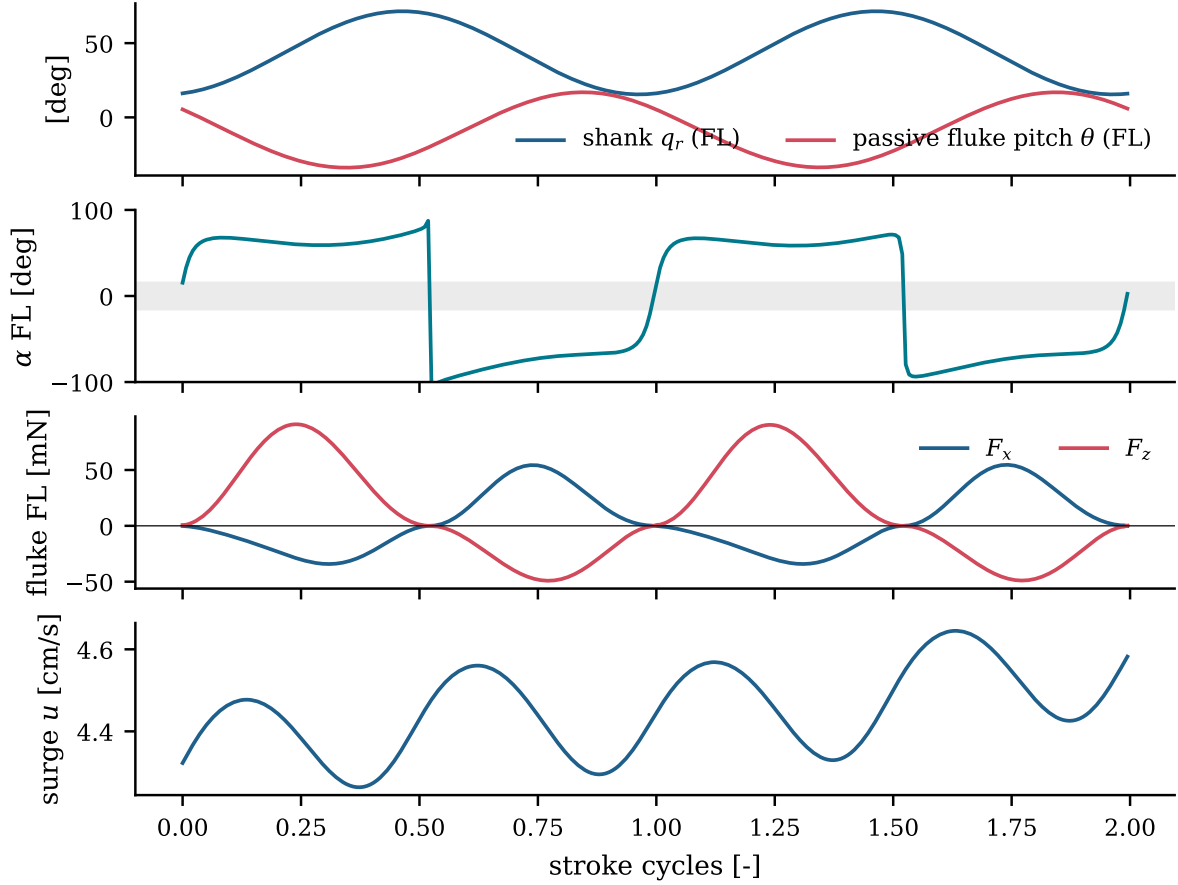


Figure 4.14: BODY2, best grid stroke ($f = 1.8 \text{ Hz}$, $q_r = 43.5^\circ \pm 28^\circ$), front-left leg over two cycles: shank angle and passive fluke pitch, angle of attack at the three-quarter chord (grey band: attached flow), fluke force components, and trunk surge speed.

4.7 Discussion and limitations

What the model can and cannot say. The reduced-order model is useful for questions whose answer depends on symmetry and kinematics: whether a parameterisation can produce thrust at all, which phase patterns cancel pitch, whether a passive hinge is needed, and where constraints bind. These conclusions hold for any reasonable coefficient values. Absolute speeds, the optimal stiffness and the optimal frequency depend directly on uncalibrated coefficients and should be treated as hypotheses for experiments.

Main modelling limitations.

- *Quasi-steady forces.* Wake, vortex shedding and the delayed-stall lift of impulsively started plates are absent. The single-leg CFD in the CFD chapter found unsteady forces about three times larger than the quasi-steady estimate for the paddle leg. Since the fluke operates almost entirely post-stall, its force level hinges on C_N and must be calibrated against CFD, UVLM or a thrust measurement at matching geometry, frequency and stiffness.
- *Isotropic added mass.* The mass trick adds the same inertia in all directions and independent of immersion. For thin paddles and flukes this overestimates the inertia along the plate and underestimates the reaction normal to it during stroke reversal.
- *No free surface.* Waves, spray and the change of immersion within an element are not

MuJoCo screenshot shot_fluke_filmstrip.png

Four frames over one stroke of the best grid gait
($f = 1.8$ Hz, $q_r = 43.5^\circ \pm 28^\circ$, diagonal), side view at water level,
frames at $t/T = 0, 0.25, 0.5, 0.75$

Figure 4.15: BODY2 swimming with the best grid stroke. The diagonal leg pairs stroke in antiphase; the flukes feather passively on the pin.

modelled beyond Equation (4.2). Strokes that leave the water are penalised only through the loss of immersion.

- *Unmodelled mechanism.* The BODY2 four-bar and servo dynamics are replaced by a position servo on the shank; backlash and velocity-dependent torque limits are missing.
- *Transient score.* The 8 s score underestimates cruising speed by a factor of about two for BODY2 (Section 4.6).

Calibration plan. In order of expected effect: (i) weigh the assembled robot with ballast and set the still-water height of O2 to -11 mm; (ii) a coast-down of the *complete* robot with legs at rest posture, fitted as in Figure 4.5, which calibrates the resistance including appendages; (iii) the mean thrust and passive pitch amplitude of a single fluke at known frequency, amplitude and stiffness, used to fit C_N and α_s ; (iv) a free-swimming run of the reference stroke, as an independent validation (target: speed within 20 %).

4.8 Summary

A robot-agnostic reduced-order simulation of paddling was built in MuJoCo from Morison elements for hull and links and a quasi-steady foil element for passively pitching flukes. It evaluates a whole-robot gait in about one to two seconds. Verification found and removed three protocol errors that had inflated earlier results. On the toy robot, the analysis showed that a first-harmonic gait cannot produce thrust with rigid flippers, that a structured 12-dimensional search outperforms the full 35-dimensional one, and that the optimiser drives the flippers into lift-based rather than drag-based propulsion. On BODY2, the diagonal pattern is the fastest and suppresses nodding by a factor of 25, the passive hinge is necessary and its best stiffness depends on frequency, and the best stroke lies on the fluke-hull clearance limit. At the robot's low advance ratio the flukes work almost entirely post-stall. Calibrating the post-stall normal force and the whole-robot resistance is therefore the next step before any predicted speed can be compared with the physical robot.

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